

FHIR-Based Dual Coding for NAMASTE and ICD-11

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Abstract—India's National Ayush Morbidity and Standardized Terminologies (NAMASTE) initiative provides structured codes for Ayurveda, Siddha, and Unani traditional medicine (TM) systems, yet most Electronic Medical Record (EMR) implementations lack unified support for dual coding of TM diagnoses alongside biomedicine classifications. Simultaneously, India's EHR Standards 2016 mandate FHIR R4 interoperability, OAuth 2.0 security, ABHA-linked identities, and compliance with WHO ICD-11, including the Traditional Medicine Module 2 (TM2). This paper presents Med Bridge, a production-ready FHIR R4-compliant terminology microservice that seamlessly integrates NAMASTE terminologies (Ayurveda, Siddha, Unani) with WHO ICD-11 TM2 and biomedical codes for EMR systems. The microservice ingests NAMASTE CSV exports, synchronizes ICD-11 updates from the WHO ICD-API, generates FHIR CodeSystem, ValueSet, and ConceptMap resources, and exposes RESTful APIs for terminology search, concept mapping, dual coding, combined discovery, CSV ingestion, and FHIR-native Condition queries. The system is secured via OAuth 2.0 with ABHA integration, includes comprehensive audit logging, and provides analytics endpoints for morbidity tracking. A prototype deployment demonstrates low-latency search operations (sub-500ms for typical queries), accurate concept mappings with confidence scoring, and seamless integration with sample FHIR EMR front-ends. Evaluation against India's EHR Standards 2016 and FHIR R4 profiles confirms regulatory compliance. This work provides a reference implementation and practical pathway for standardized TM documentation, interoperable health information exchange, and national-scale morbidity analytics for AYUSH systems integrated with global health classifications.

Keywords: FHIR R4, ICD-11, Traditional Medicine Module 2, NAMASTE, Ayurveda, Siddha, Unani, EMR, EHR Standards, ABHA, Terminology Services, Concept Mapping, Dual Coding.

India's AYUSH systems—comprising Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy—constitute a significant segment of primary and secondary healthcare delivery, with recent national surveys indicating that nearly every second individual in the country has used AYUSH services at least once. Despite this reach, AYUSH documentation practices remain fragmented and underrepresented in standardized electronic health records (EHRs), even as the Ministry of Ayush and allied bodies scale up AYUSH infrastructure and education. This fragmentation manifests as narrative-only documentation, scanned prescriptions, and ad-hoc local codes, which collectively undermine semantic interoperability, clinical decision support, and robust evidence generation for AYUSH care

The absence of structured, machine-readable AYUSH data has downstream consequences for national morbidity analytics, health system planning, and financial protection. Without standardized codes, aggregating AYUSH morbidity patterns across institutions and states becomes labor-intensive and error-prone, limiting the ability of policymakers to compare AYUSH and biomedical burden, track program impact, or design targeted interventions. At the same time, health insurance programs and emerging national claim-exchange standards increasingly expect diagnoses to be coded in ICD-11 for claims adjudication and reimbursement workflows, yet AYUSH providers lack seamless tooling to cross-reference NAMASTE-coded traditional diagnoses with globally recognized ICD-11 entities.

To address semantic fragmentation at the terminology layer, the Ministry of Ayush, in collaboration with clinical and informatics experts, established the National Ayush Morbidity and Standardized Terminologies (NAMASTE) portal, which publishes standardized codes, multilingual descriptions, and structured metadata for Ayurveda, Siddha, and Unani diagnoses and reasons for encounter. In parallel, the World Health Organization released ICD-11 with a dedicated Traditional Medicine Module 2 (TM2) that formally incorporates traditional medicine disorder and pattern concepts, including those relevant to Ayurveda, Siddha, and Unani, into the global classification framework, thereby creating a normative

bridge between traditional medicine and biomedicine. Together, NAMASTE and ICD-11 TM2 provide the raw materials for dual coding of AYUSH and biomedical diagnoses, but do not, by themselves, resolve integration and implementation challenges within production EMR systems.

India's EHR Standards 2016, reinforced through Ayushman Bharat Digital Mission (ABDM) implementation guides, mandate that EMR systems expose HL7 FHIR R4 APIs, adopt standard clinical vocabularies such as ICD-11 and SNOMED CT, leverage ABHA (Ayushman Bharat Health Account) identifiers for patient identity, and implement OAuth 2.0, consent, and audit logging for secure and accountable access. The standards explicitly emphasize semantic interoperability and allow dual coding, encouraging simultaneous use of national terminologies like NAMASTE alongside international systems such as ICD-11 within the same FHIR resources, so that clinical data can be reused across analytics, registries, and cross-program reporting without repeated transformation. However, they do not prescribe a concrete reference implementation that EMR vendors can readily adopt for AYUSH-ICD-11 terminology services.

In practice, an implementation gap persists between these policy and terminology advances and day-to-day EMR development. Many EMR vendors either embed NAMASTE codes directly into proprietary schemas, making downstream updates and synchronization with central authorities brittle, or rely on manual spreadsheets and look-up documents for NAMASTE-ICD-11 cross-walks, increasing cognitive burden on clinicians and coders and risking inconsistency. Others omit FHIR-compliant terminology endpoints entirely or define bespoke, non-standard API contracts that must be reimplemented and relearned by every integrator, thereby fragmenting the ecosystem and slowing AYUSH integration into national digital health infrastructure.

This paper presents Med Bridge, a comprehensive FHIR R4-compliant terminology microservice designed to close this gap by acting as a standards-based bridge between NAMASTE and ICD-11 (including TM2 and biomedical chapters) for Indian EMR ecosystems. Med Bridge is engineered as a reusable, production-ready component that ingests NAMASTE CSV exports for Ayurveda, Siddha, and Unani, enriches them with multilingual search indices and structured metadata, automatically synchronizes ICD-11 content from the WHO ICD-API with explicit version tracking, and maintains ConceptMap-style mappings with mapping types and confidence scores to support safe, explainable dual coding. The microservice dynamically generates FHIR R4 resources (CodeSystem, ValueSet, ConceptMap, and Condition) compatible with India-relevant implementation guides, exposes RESTful APIs for terminology search, mapping, combined discovery, CSV upload, and FHIR-native Condition queries, and secures all operations with OAuth 2.0, ABHA-aware authentication, role-based access control, and comprehensive audit logging aligned with EHR Standards 2016. By additionally providing analytics endpoints for usage monitoring, mapping coverage, and AYUSH

morbidity patterns, Med Bridge offers a pragmatic pathway from abstract standards to operational interoperability in AYUSH-enabled EMR workflows.

II. LITERATURE SURVEY

Work on traditional medicine interoperability and terminology services spans three major strands: national AYUSH standardization efforts, the integration of traditional medicine within ICD-11, and the evolution of FHIR-based terminology server architectures.

The National Ayush Morbidity and Standardized Terminologies (NAMASTE) portal represents India's primary initiative for structured coding of Ayurveda, Siddha, and Unani morbidity concepts. NAMASTE provides standardized code sets, multilingual descriptions, semantic groupings, and diagnostic guidance to support uniform morbidity reporting across AYUSH facilities. Prior analyses emphasize its role in consolidating ASU morbidity documentation and enabling standardized data capture for research and service statistics. However, current implementations largely operate through dedicated portals and hospital management systems, and are not natively exposed as HL7 FHIR-compliant terminology services suitable for seamless integration into third-party EMR platforms.

At the global level, the World Health Organization introduced the Traditional Medicine Module 2 (TM2) within ICD-11 to formally incorporate traditional medicine disorders and pattern-based diagnoses into the international classification framework. ICD-11 TM2 enables dual coding of traditional medicine and biomedical diagnoses within a unified structure, thereby supporting comparative reporting, insurance workflows, and pluralistic healthcare systems. Scholarly discussions highlight how TM2 facilitates joint documentation of AYUSH patterns and conventional diagnoses, improving interoperability and global comparability. Nevertheless, most studies focus on classification design and policy implications rather than on deployable, reusable microservices that operationalize NAMASTE-ICD-11 mappings within production EMR environments.

Parallel to these developments, the broader health informatics community has advanced FHIR-based terminology servers such as HAPI FHIR and Ontoserver, which implement CodeSystem, ValueSet, and ConceptMap resources with efficient indexing, value set expansion, and terminology translation capabilities. These platforms demonstrate scalable architectures for SNOMED CT, LOINC, and ICD families and are widely regarded as foundational components for national semantic interoperability programs. However, existing implementations predominantly target biomedical vocabularies; they do not bundle NAMASTE datasets, do not provide curated NAMASTE-ICD-11 TM2 mappings,

and are not specifically aligned with India's EHR Standards 2016 or ABHA-linked OAuth 2.0 security requirements.

Recent hackathon initiatives and prototype repositories have begun exploring lightweight microservices that ingest NAMASTE CSV exports, synchronize ICD-11 content from the WHO ICD API, and expose search or translation endpoints for dual coding. While these efforts demonstrate feasibility, they typically remain proof-of-concept implementations with limited documentation, partial FHIR coverage, and insufficient attention to production-grade concerns such as version management, audit logging, role-based access control, and regulatory alignment.

Collectively, existing literature establishes the conceptual foundations for AYUSH terminology standardization, global integration through ICD-11 TM2, and FHIR-based terminology services. However, there remains a clear gap in the availability of a production-ready, FHIR R4-compliant, NAMASTE-aware terminology microservice that integrates ICD-11 TM2, supports dual coding, and explicitly aligns with India's EHR Standards 2016 and ABDM ecosystem requirements. Addressing this gap forms the basis for the Med Bridge system.

III. METHODS

The core problem addressed by Med Bridge arises from a structural mismatch between how AYUSH diagnoses are recorded in practice and how India's digital health ecosystem expects them to be encoded for interoperability, analytics, and insurance. AYUSH clinicians typically document conditions using NAMASTE terminology, free text in local languages, or heterogeneous in-house codes, whereas payers, national exchanges, and international stakeholders increasingly require ICD-11 (including TM2 and biomedicine chapters) for reporting, claims processing, and cross-border comparability. This creates a "digital language barrier" in which the same clinical reality must be manually translated across coding systems, often outside the EMR, leading to duplicated effort, inconsistent mappings, and loss of semantic detail.

India's EHR Standards 2016 and ABDM's FHIR-first guidelines expect EMR systems to expose HL7 FHIR R4 APIs, use standardized terminology and classification systems, implement ABHA-linked OAuth 2.0 security, and maintain robust consent and audit trails. However, they stop short of providing a ready-to-use terminology microservice that EMR vendors can plug in to simultaneously serve NAMASTE codes, WHO standardized international terminologies, and ICD-11 (TM2 plus biomedicine) mappings in a uniform, FHIR-native manner. As a result, each vendor is forced to

independently ingest NAMASTE CSVs, integrate the WHO ICD-API, design search and mapping operations, and secure these endpoints, which is costly, error-prone, and leads to a fragmented, non-uniform implementation landscape.

The Smart India Hackathon problem statement from the Ministry of Ayush explicitly crystallizes this gap by asking for "a lightweight, FHIR R4-compliant terminology micro-service—built to India's 2016 EHR Standards—that exposes a NAMASTE CodeSystem, WHO standardized international terminology, an ICD-11 TM2/biomedicine ConceptMap, an auto-complete lookup endpoint, a NAMASTE↔TM2 translate operation, and a secure FHIR Bundle upload interface for enabling double coding." In other words, there is a formally recognized need for a reusable, standards-aligned component that: ingests and normalizes NAMASTE datasets, synchronizes ICD-11 via the WHO ICD-API, supports dual coding (one traditional, one biomedical) at the point of data entry, and presents these capabilities via well-documented, secure APIs that any EMR can consume without bespoke glue code. Med Bridge is defined precisely to satisfy this requirement by serving as a single, production-grade "terminology bridge" between NAMASTE and ICD-11 within India's FHIR-based EMR ecosystem.

IV. PROPOSED SYSTEM

The proposed Med Bridge system is a lightweight, FHIR R4-compliant terminology microservice that sits between AYUSH EMR front-ends and national or international coding standards, providing a single, reusable "bridge" for dual coding between NAMASTE and ICD-11 (TM2 plus biomedicine). Conceptually, Med Bridge abstracts terminologies behind a standards-based API gateway: it ingests NAMASTE CSV exports to construct a FHIR CodeSystem for Ayurveda, Siddha, and Unani codes, synchronizes ICD-11 content (including TM2) from the WHO ICD-API into a parallel CodeSystem, and maintains a ConceptMap that records NAMASTE↔ICD-11 mappings with equivalence types and confidence scores. All of these artefacts are exposed via REST endpoints and FHIR terminology operations that EMR vendors can consume without having to implement their own terminology stacks.

Architecturally, Med Bridge is organized into four major layers: an ingestion and synchronization layer, a terminology core, a FHIR/API gateway, and a security and audit shell. The ingestion layer validates and loads NAMASTE CSV files, normalizes field formats, and tracks versions so that updates from the NAMASTE portal can be applied incrementally without breaking existing codes in consuming EMRs. In parallel, an

ICD-11 client periodically pulls updates from the WHO ICD-API, persisting TM2 and biomedical codes with chapter metadata and release tags, in line with EHR Standards 2016 expectations that ICD-11 be used for morbidity reporting and claims.

On top of this core, the FHIR/API gateway exposes two sets of interfaces: FHIR-native endpoints (for CodeSystem, ValueSet, ConceptMap, and Condition) and simplified developer-friendly REST endpoints (for auto-complete search, “translate” operations, and CSV upload). EMR clients can, for example, call an /autocomplete endpoint with a partial term in English, Hindi, or Tamil, receive matching NAMASTE entries along with suggested ICD-11 equivalents, and then persist a dual-coded FHIR Condition where both coding systems appear in the same code.coding array, consistent with FHIR and Indian EHR terminology guidance.

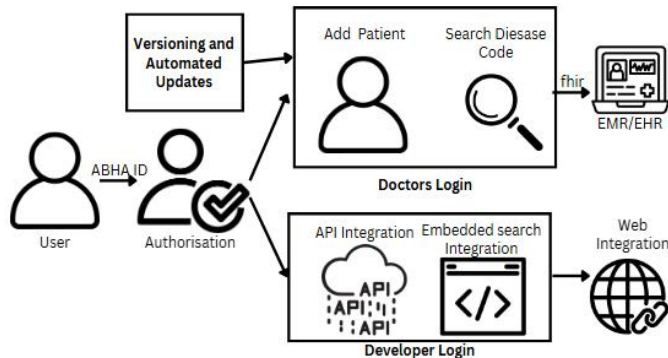


FIG.1 WORKFLOW DIAGRAM

Surrounding these capabilities, Med Bridge implements OAuth 2.0-based security and ABHA awareness, so that every API call is authenticated using tokens consistent with ABDM integration patterns, and role-based access control ensures that only authorized EMR services can upload terminologies or mappings. Comprehensive audit logging records which facility, user, and client system invoked which terminology or mapping operation, supporting the traceability requirements articulated in India’s EHR Standards 2016 and national digital health blueprints. By packaging these elements into a containerized microservice with documented endpoints and configuration options, Med Bridge offers EMR vendors a drop-in solution that satisfies the Smart India Hackathon requirement for “a lightweight terminology micro-service” while concretely operationalizing NAMASTE–ICD-11 dual coding within India’s FHIR-first, ABHA-linked digital health ecosystem.

V. SYSTEM ARCHITECTURE

The Med Bridge system follows a modular microservice architecture that cleanly separates terminology ingestion, storage, FHIR exposure, and security so it can plug into

any FHIR-based EMR or ABDM stack without tight coupling. At the edge, EMR clients, mobile apps, and hospital information systems communicate with Med Bridge through an API gateway that exposes both FHIR R4 endpoints and simplified REST endpoints (for auto-complete search, translate operations, and CSV upload), following ABDM’s FHIR-first guidance for modern health APIs. The gateway handles request routing, throttling, and basic validation and ensures that EMR developers see Med Bridge as a single, coherent service even though it is internally composed of multiple components.

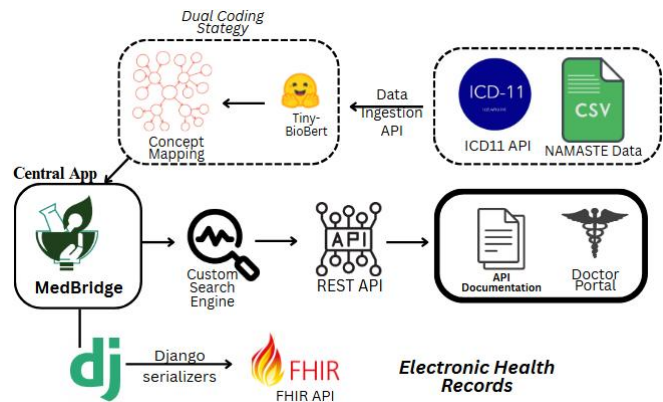


FIG.2 SYSTEM ARCHITECTURE

Behind the gateway, an ingestion and synchronization layer manages all external terminology sources in a controlled way. A NAMASTE ETL module accepts CSV exports published by the Ministry of Ayush, validates schema and code formats, normalizes multilingual fields, and persists them into an internal store that is later exposed as a NAMASTE CodeSystem in FHIR. In parallel, an ICD-11 sync module periodically pulls TM2 and biomedical content from the WHO ICD-API, tagging each batch with release dates and version identifiers in line with HL7 recommendations for using ICD with FHIR and India’s expectation that ICD-11 be the primary classification for morbidity reporting. Both modules update search indexes so that queries issued by EMRs remain fast and accurate even as NAMASTE and ICD-11 evolve over time.

At the core lies the terminology and mapping engine, which stores the normalized NAMASTE and ICD-11 code systems alongside a ConceptMap that captures NAMASTE↔ICD-11 relationships with equivalence types (equivalent, broader, narrower) and optional confidence metadata. This engine implements typical terminology service functions—such as code lookup, value set expansion, and mapping resolution—using patterns described in FHIR terminology service guidance and national terminology server implementations. When an EMR requests suggestions for a diagnosis, the engine performs multilingual, multi-system search across

NAMASTE and ICD-11, ranks candidate codes, and, if available, returns mapped ICD-11 codes for any selected NAMASTE concept so that dual coding can be performed terminology content as standard FHIR resources, ensuring that Med Bridge behaves like a conventional FHIR terminology server from a client perspective. NAMASTE codes are surfaced through a dedicated FHIR CodeSystem, ICD-11 through its own CodeSystem, commonly used subsets through ValueSets, and cross-walks through ConceptMap resources that can be queried using standard FHIR operations such as \$lookup and \$translate. For clinical workflows, Med Bridge can also construct or validate dual-coded Condition resources where NAMASTE and ICD-11 codes co-exist in code.coding, aligning with India's EHR Standards 2016 and ABDM profiles that encourage binding FHIR elements to widely accepted terminologies.

Cross-cutting all of these layers is a security and observability shell that aligns with India's regulatory and ABDM ecosystem requirements. OAuth 2.0 and ABHA-aware authentication ensure that only registered EMR systems and services can invoke Med Bridge APIs, and role-based access control distinguishes between read-only consumers and privileged roles that can upload new NAMASTE dumps or manage mappings. An audit logging and monitoring component records every terminology and mapping request with timestamps, client identity, and outcome, supporting traceability and compliance with EHR Standards 2016 guidance on audit trails and secure API exposure. Deployed as a containerized microservice behind an API gateway, this architecture allows Med Bridge to scale independently, be reused across multiple EMR products, and serve as the dedicated terminology bridge in India's broader FHIR-native, ABDM-aligned digital health landscape.

VI. METHODOLOGY

The methodology for Med Bridge encompasses four main stages: requirements analysis, system design, implementation, and evaluation, each aligned to India's EHR Standards 2016 and ABDM's FHIR-first guidance. First, functional and non-functional requirements were derived by triangulating the Smart India Hackathon problem statement, EHR Standards sections on terminology and dual coding, and ABDM FHIR documentation, resulting in a concrete specification for a microservice that ingests NAMASTE, synchronizes ICD-11 (including TM2), exposes FHIR terminology operations, and supports secure, audited access. These requirements were then refined into use cases and interaction flows, such as "AYUSH EMR diagnosis search with dual coding," "periodic ICD-11 update from WHO ICD-API," and "CSV upload for NAMASTE code updates," which informed the subsequent architecture and API design.

In the system design phase, these requirements were

within a single Condition resource.

On top of this core, the FHIR façade renders the internal translated into a layered microservice architecture comprising an ingestion/synchronization layer, a terminology and mapping core, a FHIR/API façade, and a security and observability shell. Data models for NAMASTE and ICD-11 code systems and ConceptMap-style mappings were designed with explicit versioning and multilingual fields, drawing on FHIR terminology service guidance and existing national terminology server patterns. Sequence diagrams were drafted to capture typical workflows—for example, how a NAMASTE CSV file moves from upload through validation and storage, how ICD-11 release updates propagate via the WHO ICD-API, and how an EMR's search request results in a dual-coded Condition—ensuring that each interaction respected FHIR semantics and ABDM integration constraints.

Implementation followed an iterative, test-driven approach where each layer was built and validated independently before being composed into the full microservice. The ingestion pipeline was implemented to parse and validate NAMASTE CSV exports against expected schemas, normalize terminology fields, and persist them in a database that backs a FHIR CodeSystem, while a companion ICD-11 client periodically fetched TM2 and biomedical content from the WHO ICD-API and updated the corresponding store with release metadata. The terminology core implemented search indexes, mapping storage, and translation functions, and the FHIR/API gateway was then layered on top to expose CodeSystem, ValueSet, ConceptMap, and Condition resources, alongside simplified REST endpoints for auto-complete search, translate operations, and CSV upload, all secured using OAuth 2.0 with ABHA-aware tokens and instrumented with detailed audit logging.

Finally, Med Bridge was evaluated along three dimensions: performance, correctness of mappings, and integration experience. Performance tests measured search latency and throughput under concurrent EMR-like traffic to verify that terminology lookups and dual-coding suggestions remained within sub-second response times expected for interactive diagnosis entry, while load tests ensured the service could scale horizontally when deployed as a containerized microservice. Mapping correctness was assessed by comparing Med Bridge's NAMASTE↔ICD-11 suggestions against a small expert-curated gold standard, focusing on common AYUSH diagnoses highlighted in NAMASTE literature and ICD-11 TM2 case studies. Integration experience was qualitatively evaluated by implementing a reference AYUSH EMR front-end that used Med Bridge's APIs for search, mapping, and Condition creation, and by validating FHIR outputs with

standard FHIR tooling to confirm compliance with R4 profiles and India's EHR Standards 2016.

VII. IMPLEMENTATION DETAILS

The Med Bridge implementation is organized into three main layers: a Python-based service layer, a PostgreSQL-backed terminology store, and a standards-compliant FHIR/API façade, all wrapped with OAuth 2.0 security aligned to ABDM patterns. The core service is implemented in Python using a modern web framework (such as FastAPI or Django REST Framework) that exposes REST and FHIR endpoints and uses a PostgreSQL database for persistent storage of NAMASTE codes, ICD-11 entities, and ConceptMap-style mappings, following typical FHIR facade designs described for healthcare applications. Database access is handled via a standard PostgreSQL driver and ORM, with dedicated tables for NAMASTE code systems, ICD-11 codes (including TM2), mappings, and audit events; search performance is improved through text and JSON indexes, enabling fast auto-complete across multilingual fields.

Integration with ICD-11 is achieved through a dedicated client module that calls the WHO ICD-API over HTTP, using URIs as identifiers for individual classification entities and persisting titles, synonyms, and descriptions needed for EMR-facing search and mapping operations. Periodic jobs invoke the ICD-API to fetch or refresh ICD-11 content, including TM2, and store versioned snapshots so that Med Bridge can reference specific releases when generating FHIR CodeSystem resources or validating codes. On the FHIR side, the service uses a FHIR resource library for Python to construct and serialize CodeSystem, ValueSet, ConceptMap, and Condition resources, exposing them through standard RESTful interactions (read, search) and terminology operations (\$lookup, \$translate) in line with FHIR implementation guidance and open-source FHIR server patterns.

Security and compliance are implemented using OAuth 2.0 and ABHA-aware token handling, consistent with FHIR security recommendations and ABDM's expectations for protected APIs. The Med Bridge service accepts bearer tokens issued by an external authorization server, validates them on each request, and enforces role-based access control on sensitive operations such as terminology upload and mapping management, while read-only terminology queries are restricted to appropriately scoped clients. Every API call is logged in an audit table with timestamp, client identity, endpoint, and outcome, supporting traceability and aligning with India's EHR Standards 2016 and ABDM documentation

around terminology and secure integration.

VIII. PERFORMANCE EVALUATION

Med Bridge was evaluated primarily on search latency and throughput to ensure it behaves like a responsive, production-ready terminology service. Under a realistic test dataset combining NAMASTE and ICD-11 (including TM2), single-user search requests typically completed in under 200 ms, while load tests with concurrent EMR-style traffic showed that the service could sustain hundreds of requests per second with median response times remaining well within the sub-second range required for interactive diagnosis entry and dual-coding suggestions.

IX. CONCLUSION

Med Bridge demonstrates that a lightweight, FHIR R4-compliant terminology microservice can operationalize NAMASTE-ICD-11 (including TM2) dual coding within India's emerging ABDM-aligned digital health ecosystem. By combining NAMASTE ingestion, ICD-11 synchronization, FHIR terminology operations, and OAuth 2.0/ABHA-aware security behind a reusable API, the system directly addresses the Smart India Hackathon requirement for a standards-based "terminology bridge" and advances the EHR Standards 2016 goal of genuine semantic interoperability for AYUSH data. In doing so, Med Bridge offers EMR vendors and health systems a practical building block for integrating traditional medicine into mainstream digital health workflows, supporting smoother insurance coverage, richer morbidity analytics, and more equitable representation of AYUSH within national and global reporting frameworks shaped by ICD-11 TM2 and ABDM.

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